

Ermir Mulaku

SENIOR FULL STACK ENGINEER
REACT · TYPESCRIPT · NODE.JS/NESTJS · POSTGRESQL

CONTACT

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LinkedIn
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Portfolio
www.ermir-mulaku.com

GitHub
github.com/ErmirMulaku

Location
Prishtina, Kosovo

CORE SKILLS

TypeScript & Web
TypeScript, React 19, Next.js 15/16, Redux Toolkit, TanStack Query, Angular/NgRx, SCSS/Tailwind, i18n & RTL, PWA, Core Web Vitals

Mobile & Cross-Platform
React Native / Expo (OTA, push, native printing), Electron, Socket.IO real-time

APIs & Data
NestJS, Node.js, Laravel/PHP, REST · GraphQL · gRPC; PostgreSQL in depth (EXPLAIN, locking, exclusion constraints, zero-downtime migrations); idempotency, transactional outbox, Kafka; Prisma, MySQL, queued jobs, JWT/RBAC, multi-tenant design

Elixir (project)
Phoenix, Broadway Kafka consumer

Applied AI
OpenAI function calling & embeddings, RAG chatbots, FAISS vector search, Python; Claude Code & Copilot daily

Quality & Production
Jest, Playwright, Cypress, Testing Library, Storybook, GitHub Actions CI/CD, Prometheus, health probes, feature flags, A/B testing & analytics

Cloud & Ops
AWS (App Runner, ECR, RDS), Google Cloud Run, Vercel, Docker

EDUCATION

BSc, Computer Science
University of Prishtina · 2017–2021

PROFILE

Full-stack engineer, **deeper on the frontend**: 5+ years of production **React and TypeScript** at eduwo (20,000+ users). Competent on the backend, shown in TutorHub: **Postgres correctness under concurrency**, idempotency, a **transactional outbox with Kafka** and GraphQL, proven with real-database tests. Claude Code daily, spec-first, reviewing and testing what it produces. Based in Prishtina.

EXPERIENCE

Software Engineer — Jul 2025 – Present

eduwo · Prishtina, Kosovo (Hybrid)

- Ship production **React 19 / Next.js 15** features in TypeScript on a Swiss education marketplace serving **20,000+ users** — a multi-step matching wizard, course/study/profile journeys and an AI recommendation chatbot — across four locales with SSR and PWA support.
- Built the customer-facing AI end to end: a **FAISS vector-search service** and the recommendation chatbot (service layer → Next.js API route → React UI), plus OpenAI-powered content generation in the admin.
- Design, build and document **REST APIs in Laravel** — auth, validation, queued background jobs, transactional email and notifications — consumed by the web app, the admin and native clients.
- **Own quality and uptime** for the services I ship: ESLint, Jest, Playwright E2E, pre-commit gates and monitoring of my endpoints in production.

Frontend Developer — Jul 2021 – Jul 2025

eduwo · Kosovo

- Built complex **single-page applications** with React, Redux Toolkit and TypeScript, with a hard bar on performance, responsiveness and UX detail; tracked **Core Web Vitals**.
- Ran data-driven UI experiments — **A/B testing** plus product analytics — and used production data to decide what to build and fix next.
- Designed and documented **REST API contracts** powering frontend features, coordinating payloads with backend and native consumers; built Next.js API routes as a BFF layer.
- Implemented interfaces from scratch in SCSS — responsive layouts, **animation**, carousels, charts, maps, PWA — across the modern JS toolchain (Webpack, esbuild/SWC, ESLint, Prettier, Formik + Yup).

Intern Front-End Developer — Feb 2021 – Jun 2021

Kutia · Prishtina, Kosovo

- Built responsive, accessible UI components inside a cross-functional product team, strengthening fundamentals in JavaScript, React, HTML5, CSS3 and Git workflows.

INDEPENDENT PROJECT

TutorHub — Tutoring Marketplace & Booking Backend

NestJS · PostgreSQL · GraphQL · gRPC · Kafka (Redpanda) · Elixir/Broadway · DataLoader

- Built end to end: a **NestJS API** over REST, GraphQL and gRPC, a Next.js storefront, an Angular/NgRx dashboard, an Expo app and an npm slot engine.
- Made booking **race-safe in Postgres** (conditional writes, exclusion constraints, advisory locks): **20 parallel bookings of one slot → exactly 1** (was 20).
- Made retries safe: **Idempotency-Key** across REST, GraphQL and gRPC (parallel duplicates replay one response), plus idempotent Stripe webhooks and calls.
- Moved events to a **transactional outbox → Kafka** with SKIP LOCKED relays and idempotent Node and **Elixir (Broadway)** consumers; verified end to end at lag 0.
- Tuned Postgres on a **2M-booking dataset**: deep pages **21 ms → 0.032 ms**; **zero-downtime migrations** under concurrent writes (2 ms vs a 1,968 ms write stall).
- Removed GraphQL N+1s with **DataLoaders** (150 → 3 queries for 50 bookings), added keyset pagination and errors-as-data; gated new paths behind **feature flags**.

Live: tutorhub-marketplace.vercel.app · **npm:** tutorhub-slot-engine

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SELECTED LINKS

Portfolio
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WORKING STYLE

End to end
I shape the problem with product and design, build it across the stack and follow it into production.

Production first
Instrumented, tested and observable — health probes, metrics, E2E gates, and the metrics that say whether it worked.

AI-native
Claude Code and Copilot in the loop every day; the judgement and the review stay mine.

CERTIFICATIONS

Advanced React Component Patterns
Udemy · 2022

Functional Programming in JavaScript
Udemy · 2023

Advanced JavaScript & React
Beetroot Academy · 2021

SHIPPED PRODUCTS

S2O — Cross-Platform Restaurant Platform · Viviark

NestJS · Next.js · React Native (Expo) · Electron · Socket.IO

- Built a **multi-tenant SaaS** end to end: a NestJS REST API plus a Next.js web admin, a React Native (Expo) app and an Electron desktop POS, sharing one API and a real-time Socket.IO layer.
- Built a **real-time order pipeline** (WebSocket gateways) to kitchen and bar displays with server-enforced status transitions, JWT + PIN auth and FIFO inventory with expiry tracking.
- Designed the migration from pooled to **per-restaurant isolated tenancy** (container + database per tenant) and the onboarding control plane.
- Shipped the native clients myself: **Expo OTA updates, push notifications, native printing** and Electron packaging (Windows, Linux).

  Live: App Store & Google Play — “S2O App” · s2o-menu-theta.vercel.app

Spicy Analytics — Multilingual RTL Web Platform

Next.js · i18n · RTL · SSR / SEO

- Built a production multilingual platform with **full RTL support** (English LTR + Arabic RTL, mirrored layout) via a key-based i18n system, for an enterprise client roster spanning healthcare, banking and telecom.
- Implemented per-locale routing, SSR-rendered SEO metadata and next/image optimization across a responsive, content-heavy site.

  Live: spicyanalytics.com

Three60 Music — Animated E-commerce Site

Next.js · CSS animations · E-commerce

- Built an **animation-rich**, responsive Next.js storefront with cart and CDN-served media.

  Live: three60-music.com